WHICH TOOL? MATCH WHAT THE QUESTION SAYS

Classify a system (linear, TI, causal, memoryless, stable, real)	Past paper: System classification
Block diagram with delays and gains	Past paper: Block diagram to equation
Anti-aliasing / reconstruction / sketch spectrum	Past paper: Sampling questions
Two-stage output with H(s) or H(z) and a given input	Past paper: Transfer function questions
Draw or simplify x(−t+2), x(2t), x[n−3]	W1 Transformations: reflect first, then shift
Is this signal periodic? Find T_o or N_o	W1 Periodicity (DT needs $2\pi/\omega_o$ rational)
Linear? Time-invariant? Causal? Stable? Memory?	W2 Property tests, then W3 table if you have h
Find y given x and h (or h given the equation)	W3 Convolution (flip h, then shift)
Periodic input, find output	W6: $Y_k = H(k\omega_o)X_k$
Input is a sinusoid or $e^{j\omega t}$	W5: output = $\lvert H(\omega)\rvert\cos(\omega t+\varphi+\angle H(\omega))$
Aperiodic input, signal spectrum, Parseval	W8 CTFT / DTFT
Sampling rate, aliasing, reconstruction	W9: $T_s < \pi/W$
Differential equation, RoC, poles, stability (CT)	W10–W11 Laplace, H(s)
Difference equation, RoC, poles, stability (DT)	W12 Z-transform, H(z)
Give the transform of a signal	Always state the RoC with it (Laplace, Z)
Output with initial rest or step input	Y = HX, then partial fractions, RoC outside poles if causal

TOPICS

Exam What the formula sheet gives you and what it does not

Use when: Checked against Formula_Sheet.pdf (7 pages).

Given: sifting, convolution (DT and CT), geometric sum $1/(1-\alpha)$, integrals of e^x and x^k, $X_s(\omega) = (1/T)\Sigma X((\omega-2\pi k)/T)$

Needs: **Trap:**

Given: CT and DT Fourier series tables, CTFT and DTFT tables, Laplace and Z tables with properties

Needs: Pairs and properties only. **Trap:** No RoC rules, no partial-fraction recipe, no stability tests.

Not given: system property tests, periodicity rules, aliasing condition $T_s<\pi/W$, ideal filter responses, Parseval tricks

Needs: **Trap:** These are what this sheet is for.

Laplace table has NO left-sided pair

Needs: Only right-sided signals: δ, u, tu, $e^{-at}u$, $te^{-at}u$, $e^{-bt}\cos/\sin(\omega_o t)u$. **Trap:** You must supply $-e^{-at}u(-t) \Leftrightarrow 1/(s+a)$, $\mathrm{Re}(s)<-\mathrm{Re}(a)$ yourself. Also $\cos(\omega_o t)u$ and $\sin(\omega_o t)u$ are the b=0 cases.

Z table includes $\alpha^n u[-n]$ ($|z|<|\alpha|$) but the left-sided $u[-n-1]$ form is not listed

Needs: **Trap:** Left-sided: $-\alpha^n u[-n-1] \Leftrightarrow 1/(1-\alpha z^{-1})$, $|z|<|\alpha|$. Write it from your notes.

DT Fourier series table has a typo in the square-wave row (see Fourier series card)

Needs: **Trap:** Also the DT Parseval line prints "dt". Read it as a sum.

CT Fourier series square wave: $\sin(k\omega_0 T)/(k\pi)$ for $u(t+T)-u(t-T)$

Needs: Given. **Trap:** $X_o = 2T/T_o$ is not listed. Take the limit $k\to 0$ of the formula.

Exam System classification

Use when: Seen: 2022 Q2, 2024 Q1, mid-sem Q1, sample and mock.

Memoryless? Only the current input appears, with the same time index

Needs: **Trap:** x(t+2), x[n−1], sums over k all have memory.

Causal? Any input at a later time than the output?

Needs: Memoryless ⇒ causal. **Trap:** Σ over k=−1..1 of x[n−k] uses x[n+1], so non-causal. $e^{-t}x(t+2)$ is non-causal.

Linear? Check $x \to 0$ gives 0, then scaling and addition

Needs: A fixed multiplier of the signal is fine (e^{-t}, sin t, n^2). **Trap:** Squaring x, adding a constant, or |x| all break linearity. Initial rest keeps y=x+0.5x(t−1) linear.

Time-invariant? Does any coefficient depend on t or n?

Needs: Time-varying multiplier ⇒ TV. **Trap:** $e^{-t}x(t+2)$ is TV. $2e^{j\omega_1 t}x(t-1)$ is TV. $x[n-1]^2+x[n]$ is TI but non-linear.

BIBO stable? Bound |y| using $|x| \le M$

Needs: Finite sum of delayed copies ⇒ stable. **Trap:** Complex exponential multipliers have magnitude 1, so they stay stable.

Real system? Real input must always give real output

Needs: Complex coefficients or $e^{j\omega t}$ multipliers ⇒ not real. **Trap:** 2024 Q1: $y = 2e^{j\omega_1 t}x(t-1)+e^{j\omega_2 t}x(t-2)$ is linear, TV, causal, has memory, stable, NOT real.

"Is it LTI?" for a system that creates new frequencies

Needs: An LTI system only scales and phase-shifts each input frequency. **Trap:** Mock Q6d: 440 Hz tone in, 330 and 550 Hz out ⇒ not LTI.

Stuck? Try: Write y for x=0, for 2x, and for x_1+x_2, then for a delayed x. Four lines, four answers.

Exam Sampled values and DT sequences

Use when: Seen: 2022 Q1, 2024 Q2.

$x_s[n] = x(nT_s)$. Build a table of $t = nT_s$ first

Needs: Fill n=0..5 in order. **Trap:** cos(2πt) at 10/s: 1, 0.81, 0.31, −0.31, −0.81, −1.

sin(10πt) at 10 samples/s

Needs: $\omega_o T_s = \pi$. **Trap:** All samples are 0 (critical sampling).

u(t−0.25) at t=0,0.1,...,0.5

Needs: **Trap:** 0, 0, 0, 1, 1, 1.

sin(2πt)u(t) at 10/s

Needs: **Trap:** 0, 0.59, 0.95, 0.95, 0.59, 0.

$z(t)=u(t+1)-u(t-1)$, $T_s=2/3$: $x[n]=1$ for $|nT_s| < 1$, i.e. $n = -1, 0, 1$

Needs: Evaluate at nT_s, n = −2, −1, 0, 1 gives 0, 1, 1, 1. **Trap:** Then $y[n] = \Sigma_{k=-1}^{1} h[n-k]$: A=−1, B=1.

Stuck? Try: Draw the CT signal and mark the sample instants nT_s on the axis.

Exam Block diagram to difference equation

Use when: Seen: 2024 Q2c, mock Q2, sample Q2.

Label each wire with its Z-transform. A delay multiplies by z^{-1}

Needs: Do the algebra in the z-domain. **Trap:** Gains multiply the wire they sit on. Adders add the arriving wires.

Collect: $\hat{Y} = \Sigma(\text{gain} \times \text{delays})\ \hat{X} + \text{feedback} \times \hat{Y}$, then read each coefficient

Needs: **Trap:** Read the shared final gain (0.1 in 2024 Q2c) into every term.

2024 Q2c: $y[n]=0.5y[n-1]+x[n]+0.2x[n-1]+0.1x[n-2]$

Needs: Matching coefficients gives a=1, b=2, d=0.5 with c=10 and final gain 0.1. **Trap:**

Direct difference equation to FIR diagram: $y[n] = \Sigma\ b_k\ x[n-k]$

Needs: Delay chain, tap gains b_k, one adder. **Trap:** $h[n] = \Sigma\ b_k\ \delta[n-k]$ (mock Q1).

Exam Read h or the transform from a given expression

Use when: Seen: sample Q4, mock Q1, mid-sem (impulse response of deltas).

Impulse response of an FIR equation: replace x by δ

Needs: LTI. **Trap:** $h[n] = \delta[n] - 0.5\delta[n-1] - 0.5\delta[n-2] + \delta[n-3]$.

$H(\omega) = 1.5 + 0.5\cos\omega + \cos 2\omega$: rewrite with Euler, then the coefficient of $e^{-j\omega n}$ is $h[n]$

Needs: $\cos\omega = \frac{1}{2}e^{j\omega} + \frac{1}{2}e^{-j\omega}$. **Trap:** $h[-2]=0.5$, $h[-1]=0.25$, $h[0]=1.5$, $h[1]=0.25$, $h[2]=0.5$. Not causal.

If $H(\omega)$ is multiplied by $e^{-j\omega N}$, the whole h shifts right by N

Needs: Delay property. **Trap:** Multiply by $e^{-j2\omega}$: all indices ≥ 0, so the system becomes causal.

$h(t) = \delta(t-1) - \delta(t-2) + \delta(t-3)$

Needs: Finite number of impulses ⇒ $\int|h| = 3 < \infty$. **Trap:** Stable. Step response $s(t) = u(t-1) - u(t-2) + u(t-3)$. Output = $x(t-1) - x(t-2) + x(t-3)$.

Sketch the output of a delta-sum system for a pulse input

Needs: Delay the pulse by 1, 2, 3 and add with signs. **Trap:** The copies overlap: add them point by point.

Exam Fourier series questions

Use when: Seen: 2022 Q3, 2024 Q4, mid-sem Q2, sample Q8, mock Q3 and Q4.

$X_0 = (1/T_0)\int$ over one period of x = average value

Needs: Periodic extension of a finite pulse. **Trap:** 2022 Q3: $x=1-t^2$ on $|t|\le1$, $T_o=2$ ⇒ $X_o = 2/3$. 2024 Q4: duty cycle β ⇒ $X_o = \beta$.

$Y_k = H(k\omega_0)\ X_k$. $Y_0 = H(0)\ X_0$

Needs: Stable LTI, periodic input. **Trap:** 2022 Q3c: $h = e^{-2t}u - e^{-4t}u$ ⇒ $H(0)=1/2-1/4=1/4$, $Y_o=1/6$.

Parseval: $(1/T_0)\int|z|^2 = \Sigma_{k=-\infty}^{\infty}|Z_k|^2$. Real z: $\Sigma_{k=1}^{\infty}|Z_k|^2 = (\text{power} - |Z_0|^2)/2$

Needs: Real, periodic. **Trap:** Mid-sem: power 4, $Z_o=0$ ⇒ $\Sigma_{k\ge1}|Z_k|^2 = 2$.

Which harmonics survive a filter? Find $k\omega_0$ for each k and compare with the passband

Needs: Include negative k. **Trap:** Sample Q8: $T=2\pi$ ⇒ only $k=\pm1$ passes ⇒ $y = 2|X_1|\cos t$. $T=4\pi$ ⇒ $k=\pm2$ passes. $T=8\pi$ ⇒ several pairs ⇒ not a single cosine.

Real pair X_k, X_{-k} both in the passband ⇒ $A\cos(k\omega_0 t + \angle X_k)$ with $A = 2|X_k|$

Needs: Real signal. **Trap:**

Modulation: $\cos(\omega_c t)x$ ⇒ $W_k = \frac{1}{2}X_{k-1} + \frac{1}{2}X_{k+1}$ when $\omega_c = \omega_0$

Needs: ω_c is an integer multiple of ω_o. **Trap:** Demodulator (mock Q4): choose A so $Y_o = A \cdot W_o$ matches, and $0 < B < \omega_c$ to block $k \neq 0$. Linear but not time-invariant.

Fundamental frequency of a sum of sinusoids = gcd of the component frequencies

Needs: **Trap:** Mock Q6a: 440, 330, 550 Hz ⇒ 110 Hz, so $\omega_o = 220\pi$.

Differential equation from $H(\omega)$: cross-multiply, then $j\omega \to d/dt$

Needs: Initial rest. **Trap:** 2024 Q4a: $(1+ja\omega-\omega^2)Y = (1-(\omega/2\pi)^2)X$ ⇒ $y + a\,y' + y'' = x + x''/4\pi^2$.

If $H(k\omega_0)=0$ at some harmonic, x is not unique from y

Needs: Add or remove that harmonic. **Trap:** 2024 Q4d: add $10\cos(2\pi t+\varphi)$ to x when $H(2\pi)=0$.

Stuck? Try: Write ω_o, list which k are relevant, evaluate $H(k\omega_o)$ for each. Then build y.

Exam Frequency response and sinusoid questions

Use when: Seen: 2022 Q4, sample Q6, mock Q6.

Output sinusoid → input: divide the output phasor by $H(\omega_0)$

Needs: Real stable system. **Trap:** Do each frequency separately.

Input constant a ⇒ output $a \cdot H(0)$

Needs: $x = a\,e^{j0t}$. **Trap:** Sample Q6: $H(\omega)=Y/X$, then $y = a \cdot H(0) = 1$.

Periodic input: $Y_k = H(k\omega_0)X_k$

Needs: **Trap:** Use $\omega_o = 2\pi/T_o$.

Delay + gain system: $h = 0.5\delta(t-0.1)$, $H(\omega) = 0.5e^{-0.1j\omega}$

Needs: **Trap:** Undo it: $z(t) = 2y(t+0.1)$.

$H = Y/X$ using FT tables

Needs: **Trap:** If $X(\omega) = 0$ anywhere, H is undetermined there.

Exam Fourier transform algebra

Use when: Seen: 2022 Q5, 2024 Q7, mock Q3.

Improper rational $(4+j\omega)/(5+j\omega)$: divide first $= 1 - 1/(5+j\omega)$

Needs: Degree numerator = degree denominator. **Trap:** The constant 1 gives $\delta(t)$. $y(t) = \delta(t) - e^{-5t}u(t)$.

$X = Y/H$, then simplify and look up each term

Needs: $H(\omega) \neq 0$. **Trap:** 2024 Q7: $X = (2+j\omega)/(5+j\omega) = 1 - 3/(5+j\omega) \Rightarrow x = \delta(t) - 3e^{-5t}u(t)$.

$n\alpha^n u[n] = (n+1)\alpha^n u[n] - \alpha^n u[n] \Rightarrow 1/(1-\alpha e^{-j\omega})^2 - 1/(1-\alpha e^{-j\omega})$

Needs: Table has $(n+1)\alpha^n u[n]$ and $\alpha^n u[n]$. **Trap:** 2022 Q5a with $\alpha=0.2$.

$0.2^n \cos(n)\, u[n]$: write $\cos n = \tfrac{1}{2}e^{jn} + \tfrac{1}{2}e^{-jn}$, then modulation

Needs: Frequency shift by ± 1. **Trap:** $\tfrac{1}{2}/(1-0.2e^{-j(\omega-1)}) + \tfrac{1}{2}/(1-0.2e^{-j(\omega+1)})$.

$\cos(\omega_0 t)x(t) \Rightarrow \tfrac{1}{2}X(\omega-\omega_0) + \tfrac{1}{2}X(\omega+\omega_0)$

Needs: **Trap:** Mock Q3: $\cos(a\pi t)x$ with 10 periods in length 2 $\Rightarrow a = 10$, $X(\omega)=2\sin\omega/\omega$.

Use the table you are given

Needs: Current formula sheet: $u[n+N]-u[n-N] \Leftrightarrow [\sin(N\omega)/\sin(\omega/2)]e^{j\omega/2}$ (2N samples, n=−N..N−1). **Trap:** The 2024 paper printed $\sin(\omega(N+\tfrac{1}{2}))/\sin(\omega/2)$, which is the (2N+1)-sample window $u[n+N]-u[n-N-1]$. Count the samples before choosing.

Exam Sampling, aliasing, reconstruction questions

Use when: Seen: 2022 Q6, 2024 Q6 and Q10 (sample), mock Q8, sample Q1.

No aliasing iff sampling frequency $>$ 2 × highest frequency ($T_s \cdot W < \pi$)

Needs: Sinusoid: $\sin(4\pi t)$ has f=2 Hz, so $f_s>4$. **Trap:** Bandpass signal: use the highest edge W, not the band width. 5π–$8\pi \Rightarrow W=8\pi \Rightarrow$ Nyquist rate 8 samples/s.

Anti-aliasing LPF cutoff $\leq f_s/2$ Hz $= \pi f_s$ rad/s

Needs: Ideal LPF $h = \sin(Wt)/(\pi t)$ has $H = u(\omega+W) - u(\omega-W)$, cutoff W. **Trap:** Sample Q1: largest $W = \pi f_s$. $f_s=500 \Rightarrow 500\pi$ rad/s.

Hz to rad/sample: $\omega = 2\pi f/f_s$

Needs: DT filter design. **Trap:** Sample Q10: 50 Hz at $f_s=500 \Rightarrow \pi/5$. High-pass edge at $\pi/5$. Sketch -2π to 2π with edges $\pm\pi/5$, $\pm(2\pi-\pi/5)$.

$X_s(\omega) = (1/T_s)\,\Sigma\, X((\omega-2\pi\ell)/T_s)$. Inverse: $X(\omega) = T_s\, X_s(T_s\omega)$ inside the band

Needs: Mock Q8: peak 20 → X peak 2, edge $(2\pi/3)/0.1 = 20\pi/3$. **Trap:** Horizontal axis scales by $1/T_s$, vertical by T_s.

Energy of x_s: $\Sigma|x_s[n]|^2 = (1/2\pi)\int$ over one period of $|X_s|^2$

Needs: Parseval. **Trap:** Mock Q8b = 800/9. Use evenness to integrate over 0 to π and double.

Impulse-train sampling $p(t)=\Sigma\delta(t-nT)$: $Z(\omega) = (1/T)\,\Sigma\, X(\omega-2\pi k/T)$

Needs: $T = 1/4 \Rightarrow \omega_s = 8\pi$, copies every 8π. **Trap:** 2024 Q6: $H_2 = T = 1/4$ on the band, 0 elsewhere. $H_1 = 1$ or a band-limiting filter.

ZOH: output is held; reconstruction delay $\approx T_s/2$

Needs: **Trap:** Order of ideal filter and sampler matters only if the signal is not already band-limited.

Always say why: "sampled at f_s above twice the highest frequency, so the DTFT copies do not overlap"

Needs: **Trap:** Marks are for the sentence.

Exam Transfer function, RoC, stability, output

Use when: Seen: 2022 Q7, 2024 Q8, mock Q5 and Q7, sample Q5.

Block diagram: name wires, write equations in s, solve for $\hat{Y}/\hat{X}$

Needs: Integrator block 1/s, delay e^{-sT}, gain constants. **Trap:** 2024 Q8: $\hat{H} = (2s^2-1)/((s-1)(s+2))$.

RoC: "all subsystems causal" ⇒ right of the rightmost pole

Needs: Rational $\hat{H}$. **Trap:** 2024 Q8: poles 1 and −2 ⇒ RoC Re(s) > 1.

Stable? RoC must contain the jω axis; equivalently all poles in the open left half-plane

Needs: **Trap:** Pole at +1 ⇒ not stable. Poles at ±2j (sample Q5) ⇒ not stable.

Output for a given input: $\hat{Y} = \hat{H}\hat{X}$, partial fractions, inverse with the right-sided table

Needs: Causal system, right-sided signals. **Trap:** $te^{-t}u + e^{-t}u$ has $\hat{X} = (s+2)/(s+1)^2$. It cancels the (s+2) pole.

Repeated pole at −1: $\hat{Y} = A/(s+1)^2 + B/(s+1) + C/(s-1)$

Needs: Get C by cover-up, A by cover-up, B by evaluating at s=0 (or any s). **Trap:** 2024 Q8e: $y = -\tfrac{1}{2}te^{-t}u + (7/4)e^{-t}u + (1/4)e^{t}u$.

Step response: inverse of $\hat{H}(s)/s$

Needs: Causal. **Trap:** Sample Q5: $\hat{H} = 2s/(s^2+4)+s/(s+1) \Rightarrow y = [\sin 2t + e^{-t}]u(t)$.

Bounded step response does NOT imply BIBO stability

Needs: **Trap:** Same example: bounded step response, poles on the jω axis.

ODE $y''+ay'+by=x \Rightarrow \hat{H} = 1/(s^2+as+b)$. Causal and stable ⇒ poles with negative real part

Needs: **Trap:** Pick p=−1, q=−2 ⇒ $s^2+3s+2 \Rightarrow a=3$, b=2.

$\hat{Y}=1/(s(s-1))$, RoC Re(s)>1 ⇒ $y = (e^t-1)u(t)$

Needs: Right-sided inverse for both terms. **Trap:** $\hat{H} = \hat{Y}/\hat{X} = (s+2)/(s(s-1))$ when $\hat{X}=1/(s+2)$ (RoC Re(s)>−2).

Exam Poles, zeros and properties puzzle

Use when: Seen: sample Q3. Every property in the statement is a clue.

"Input 1 for all t gives output 0" ⇒ $H(0) = 0$ ⇒ zero at $s = 0$

Needs: Constant = e^{0t}. **Trap:**

"Real-valued system" ⇒ complex poles and zeros come in conjugate pairs

Needs: **Trap:** Pole at −1+j ⇒ also −1−j.

"RoC is Re(s) > 2" ⇒ rightmost pole has real part 2

Needs: **Trap:** Three poles, two used by the pair, so the third is real: s = 2.

Right-sided RoC ⇒ right-sided impulse response. Strip $-1<\mathrm{Re}(s)<2$ ⇒ two-sided

Needs: **Trap:**

Write each answer with the property that forced it

Needs: The hint says: refer to all properties. **Trap:** 1 mark for the location, 1 for the reason.

Exam **Z-domain questions**

Use when: Seen: 2022 Q8, 2024 Q3 and Q5, mock Q9, sample Q7.

$y[n]+2y[n-1]+y[n-2] = \frac{1}{2}x[n]+x[n-1]-x[n-2]$: $\hat{H} = (\frac{1}{2}+z^{-1}-z^{-2})/(1+2z^{-1}+z^{-2})$. Multiply by $2z^2$: $(z^2+2z-2)/(2z^2+4z+2)$

Needs: To match the form $(z^2+az+b)/(cz^2+dz+e)$: a=2, b=−2, c=2, d=4, e=2.
Trap: Initial rest ⇒ causal and LTI.

Poles: solve $z^2+z+1=0 \Rightarrow z = -\frac{1}{2} \pm j\sqrt{3}/2 = e^{\pm j2\pi/3}$

Needs: Polar form means magnitude and angle. **Trap:** |z|=1: on the unit circle.

Causal ⇒ RoC $|z| > 1$. Unit circle not inside ⇒ not BIBO stable

Needs: **Trap:** Poles on the circle are not stable.

h[n] for $(1+0.5z^{-1})/(1+z^{-1}+z^{-2})$: match $\alpha^n\cos(\omega_0 n)u[n]$ with $\alpha=1$, $\omega_0=2\pi/3$ ($-\cos\omega_0 = 0.5$)

Needs: **Trap:** h[n] = cos(2πn/3)u[n].

Unbounded output for a bounded input: a real bounded x at the pole frequency

Needs: Resonance. **Trap:** x[n]=cos(2πn/3) ⇒ y grows. Say "bounded by 1, output unbounded".

Given $x[n]=(0.5)^n u[n]$, $y[n]=(0.25)^n u[n]$: $\hat{H} = \hat{Y}/\hat{X} = (1-0.5z^{-1})/(1-0.25z^{-1})$

Needs: Table pairs for $\alpha^n u[n]$. **Trap:** IIR (pole at 0.25 ≠ 0). h = 2δ[n] − $0.25^n u[n]$.

FIR ⇔ no poles except z=0. IIR ⇔ a pole elsewhere

Needs: **Trap:**

Step response in Z: $\hat{H}(z)/(1-z^{-1})$, RoC $|z|>1$

Needs: Causal. **Trap:** Sample Q7: $1/((1+z^{-1}+z^{-2})(1-z^{-1})) = 1/(1-z^{-3})$.

Write the difference equation from $\hat{H}$: cross-multiply, $z^{-m} \rightarrow$ delay by m

Needs: State "subject to initial rest". **Trap:** Missing initial rest loses a mark.

2022 Q8: $y[n]=x[n]+2x[n+1]-2y[n-1]$: x[n+1] gives a positive power of z in the numerator

Needs: Pole at −2, so RoC $|z|>2$ for a causal solution. **Trap:** Not stable (pole outside the unit circle).

Stuck? Try: Always: transfer function → poles → RoC from causality → stability test → output.

Quick version: past-paper question types only. The full sheet has the week-by-week reference.